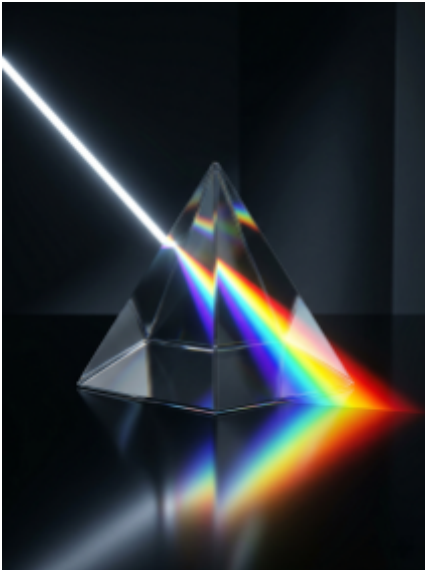


EBA5008: Graduate Certificate in Intelligent Financial Risk Management

The Invisible Applicant: Illuminating the Journey from Data Desert to Credit Worthiness

A Fairness-Aware Risk Engine and Gateway for Thin-File and Credit-Invisible Borrowers



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Team Name

Sovereign Prism

Team Members

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1. Introduction

No File? No Problem! At Compliance Vanguard, our aim is to broaden inclusive lending for underserved borrowers without a safety net, by leveraging alternative data and baking in fairness and accuracy into our compliance-first algorithms. In this way, we aim to Crack the Credit Gap for The Invisible Applicant.

Traditional credit scoring relies heavily on bureau data from agencies such as TransUnion, Equifax, and Experian. While effective for consumers with established credit histories, this approach systematically excludes approximately 26 million Americans who are "credit invisible" and 19 million more with "thin files" - groups that disproportionately include minority communities, younger consumers, and recent immigrants.

Non-traditional data sources - including digital footprints, phone usage patterns, social network risk signals, identity verification depth, and external scores; offer the potential to score previously unscorable consumers. However, their adoption raises critical questions about algorithmic fairness, regulatory compliance, and model interpretability.

This project investigates whether alternative data features can improve credit default prediction accuracy; particularly for thin-file applicants; while maintaining algorithmic fairness across gender and socioeconomic groups, and satisfying Basel III, IFRS 9, and stress testing requirements. The aim is to design and develop a Risk Modelling tool for the home credit sector.

2. Business Problem & Objectives

Business Problem

Credit scorecards are mathematical formulae that assign points to applicant characteristics to predict future default behaviour. Traditional scorecards favour closeness (5–15 interpretable variables); however, millions of creditworthy consumers are excluded due to insufficient bureau data. Additionally, as models incorporate richer feature sets, the risk of proxy discrimination increases - features that appear neutral (e.g., phone stability) may correlate with protected attributes such as age, gender, or socioeconomic status.

Business Objectives

This project builds a credit risk model that responsibly expands access to formal credit for applicants who are invisible to traditional bureau-based scoring. Leveraging all seven Home Credit relational tables - linking application data with bureau records, payment histories, and behavioural signals - we aim to achieve three outcomes: a 3–5 percentage point AUC-ROC improvement over a bureau-only baseline, reliable risk ranking for 60–80% of applicants who would otherwise go unassessed, and fairness metrics within defensible bounds (disparate impact ratios ≥ 0.80 , equalized odds differences < 0.05) so that lending decisions remain explainable and compliant.

Technical Objectives

The project constructs an end-to-end pipeline from raw ingestion of all seven Home Credit tables through feature engineering (~250 applicant-level features), model training, fairness assessment,

regulatory quantification, and monitoring. LightGBM serves as the primary model with XGBoost as comparator, evaluated across four controlled experiments: traditional-only baseline, traditional-plus-alternative integration, Fairlearn-constrained optimisation targeting gender/age/income fairness, and thin-file subgroup analysis. Regulatory outputs include Basel III risk-weighted assets and IFRS 9 expected credit loss provisions derived from model-predicted default probabilities, complemented by SR 11-7 compliant monitoring covering PSI, AUC/Gini drift, and fairness stability.

Business Impact & Value

The modelling framework enhances risk differentiation for thin-file and credit-invisible applicants, optimizing Basel III risk-weighted assets and IFRS 9 expected credit loss provisioning while expanding credit access. Fairness-constrained optimization ensures ECOA/Regulation B compliance, reducing disparate impact litigation risk. Regulatory analytics support CCAR/DFAST stress testing requirements. SR 11-7 aligned monitoring maintains model governance and performance stability over time. The approach delivers measurable improvements in portfolio risk management, capital efficiency, and regulatory compliance.

Ultimately, Sovereign Prism uses fairness-aware AI to reveal the true creditworthiness hidden within 'invisible' financial profiles. We don't just box applicants, we illuminate their potential, just like a prism that reveals the hidden colors within a seemingly singular beam of light.

3. Project Scope and Design

Dataset

The primary dataset is the Home Credit Default Risk competition dataset from Kaggle, comprising real-world consumer lending data with 7 relational tables linked by applicant and previous application identifiers.

Dataset Structure and Data Fields

Table	Rows	Columns	Content
application_train.csv	307,511	122	Applicant demographics, loan terms, TARGET label
application_test.csv	48,744	121	Holdout evaluation set
bureau.csv	1,716,428	17	External credit bureau records
bureau_balance.csv	27,299,925	3	Monthly bureau status history
previous_application.csv	1,670,214	37	Prior Home Credit applications
POS_CASH_balance.csv	10,001,358	8	POS/cash loan balances
credit_card_balance.csv	3,840,312	23	Credit card account snapshots
installments_payments.csv	13,605,401	8	Instalment payment records

4. Methodology & Architecture

The project follows a structured end-to-end pipeline from raw data ingestion through to regulatory capital output, with fairness evaluation embedded at every decision point.

Stage	Component	Output
Data Ingestion & Exclusions	Load 7 relational CSVs; apply scoring/performance exclusions; G/B/I waterfall classification	Clean training set
Feature Engineering	Aggregate bureau/payment/balance tables to applicant level; classify Traditional vs Alternative features	Feature matrix (~250 features)
Dimension Reduction	PCA / Factor Analysis (Varimax rotation) to identify latent financial dimensions; select 1–2 raw variables per factor	Shortlisted variables (~40–60)
Feature Selection	WoE binning + Information Value scoring; VIF multicollinearity check	Final feature set (15–30 variables)
Model Training	LightGBM primary; XGBoost & Logistic Regression as comparisons; stratified 5-fold CV and other models	4 trained A-Score variants + B/C-Scores
Fairness Evaluation	Fairlearn MetricFrame on gender, age, income groups; compute DI ratios, equalized odds	Fairness metrics per group
Fairness Mitigation	Threshold Optimizer (post-processing); Exponentiated Gradient (in-processing)	Fairness-constrained model
Explainability	SHAP TreeExplainer; adverse action reason code generation per ECOA/Reg B	Feature importance, reason codes
Regulatory Output	Basel III PD/LGD/EAD -> RWA; IFRS 9 ECL staging; CCAR/DFAST stress testing	Capital, provision, stress results
Model Monitoring	PSI, AUC stability, Gini drift across time-based validation splits	Monitoring dashboard metrics

4. Key Deliverables

EDA & G/B/I Analysis: The project will begin with a thorough exploratory analysis of all seven relational tables, producing visualisations of class imbalance (8.1% default rate), missing value distributions (50+ columns with >30% missing), and demographic breakdowns. A waterfall classification will be applied to derive Good, Bad, and Indeterminate labels from bureau_balance STATUS codes, with the indeterminate band excluded from model training to sharpen Good/Bad separation.

Feature Engineering Pipeline: Modular Python implementation generating applicant-level feature matrix (307,511×250) with explicit traditional/alternative partitioning (~120 traditional, ~130 alternative features), thin-file segmentation logic (1,700 credit-invisible + 50,000 thin-file applicants), and comprehensive documentation of transformations, imputation strategies (50+ columns >30% missing), and outlier handling.

Model Artifacts and Experimental Framework: 8 serialized LightGBM/XGBoost models across four experiments, stratified 5-fold CV performance tables (target: baseline AUC 0.74-0.76 → full AUC 0.78-0.80, Gini uplift 6-10pp), ROC/PR curve visualizations, feature importance rankings.

Fairness Assessment Suite: Fairlearn MetricFrame reports across gender×age×income intersections (3×5×5 combinations), mitigation analysis (pre-/in-/post-processing) comparing baseline vs. constrained models ($DI \geq 0.80$, equalized odds < 0.05 , max AUC degradation $< 1pp$), accuracy-fairness Pareto frontiers.

Regulatory Analytics Modules: Basel III IRB RWA calculator (Vasicek formula, portfolio + 4 thin-file segments), IFRS 9 ECL engine (Stage 1: 12m PD×LGD×EAD; Stage 2/3: lifetime across 307k applicants), CCAR stress scenario generator (PD×3.5/LGD×1.6 severe case).

Explainability Toolkit: SHAP TreeExplainer implementation generating portfolio-level feature importance (top-20 features) and applicant-specific adverse action reason codes (top-4 negative SHAP values mapped to ECOA categories) for 10,000-sample validation cohort.

Model Monitoring Framework: Module computing PSI (< 0.10), AUC/Gini/KS drift, fairness stability ($\Delta DI < 0.02$) across 4 time-based validation splits (quarterly windows via DAYS_REGISTRATION), configurable green/amber/red alerting thresholds for 15 protected group combinations.

5. Effort Estimates and Timeline

Below is a planned Gantt Chart emphasizing the project timelines. All = All team members; K = Kaushik; V = Venkatesh; SU = Sandesh; SS = Samarth; N = Norbert				
Activity	Plan Start	Man Days	Story Points	Calendar Duration (in Days)
Further EDA	24-Mar	8	8	6
Feature Engineering & Selection	30-Mar	6	5	4
Model Training & Tuning	3-Apr	5	5	5
Fairness Evaluation	8-Apr	5	8	3
Explainability & Interpretability	11-Apr	6	10	4
Final Evaluation & Comparison	15-Apr	4	6	2
Deployment & Test	17-Apr	8	10	5
Presentation Rehearsal	22-Apr	3	3	1
Report Generation	23-Apr	5	10	2
Total =		50.000	65.000	

Gantt Chart (Estimation Timeline)

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Activity	Mar	24	25	26	27	28	29	30	31	Apr	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
Further EDA	All	All	All	All	All	All																													
Feature Engg. & Selection										K,V	K,V	SU	SS																						
Model Training & Tuning													N	K,N	V,N	SU,SSSU,V																			
Fairness Evaluation																		K	S	SU,V															
Explainability & Interpretability																					SS	SU	K,V	K,N											
Final Evaln. & Comparison																									V	SU,N,K									
Deployment & Test																												All	All	All	All	All			
Presentation Rehearsal																																All			
Report Generation																																	All	All	

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